VERIFIED PEPTIDES NETWORK

Peptide Reconstitution Guide

Detailed Per-Compound Reconstitution Instructions · Exact Solvents · Volume & Concentration Math · Storage Protocols

STEP-BY-STEP FOR EVERY MAJOR RESEARCH PEPTIDE · ACCURATE · SEARCHABLE

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How to Use This Guide

Each peptide entry in this guide includes: the common vial sizes you will encounter, the correct solvent to use, a numbered step-by-step reconstitution protocol specific to that peptide, a concentration reference table so you can calculate your draw volumes, storage instructions, and any warnings or special handling notes.

Peptides that require acetic acid (IGF-1, LR3, MGF, Hexarelin) are flagged in red. Always read the warning section before reconstituting an unfamiliar compound.

■ **All compounds are for research purposes only. Not for human consumption.**

SOLVENT ABBREVIATION	FULL NAME	WHEN USED
BAC Water	Bacteriostatic Water for Injection (0.9% benzyl alcohol)	Default for most peptides. All multi-dose vials.
SW	Sterile Water for Injection (no preservative)	Single-dose only. Use immediately and discard.
0.6% AA	0.6% Acetic Acid in sterile water	IGF-1 family, MGF, Hexarelin. First solvent step only.
NS	Normal Saline (0.9% NaCl) for injection	IV dilution solvent. Not for SC reconstitution.

HEALING & RECOVERY PEPTIDES

BPC-157 — Body Protection Compound 157

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg, 10 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Remove vial from refrigerator. Allow to reach room temperature (15–20 min). Do not open while cold.
2	Wipe the rubber stopper of both the BPC-157 vial and BAC water vial with a fresh alcohol wipe. Allow to air-dry 30 seconds.
3	Using a 3 mL syringe, draw 2 mL of BAC water (for a 5 mg vial — see concentration table for other volumes).
4	Insert the needle into the BPC-157 vial at a slight angle, tip touching the inside glass wall.
5	Depress the plunger very slowly — let BAC water run down the glass wall and pool over the powder. Do NOT spray directly onto powder.
6	Remove syringe. Do not shake. Gently swirl or roll the vial between palms for 60–90 seconds.
7	The solution should become completely clear and colorless within 2–5 minutes. If not, roll gently for another 2 min.
8	Label vial: 'BPC-157 — [date] — [concentration] mg/mL'. Refrigerate immediately.
9	Use a new insulin syringe and wipe the stopper with a fresh alcohol wipe before each draw.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 250 mcg dose
5 mg	1 mL	5 mg/mL (5000 mcg/mL)	0.05 mL (5 units on U-100 syringe)
5 mg	2 mL	2.5 mg/mL (2500 mcg/mL)	0.10 mL (10 units)
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.25 mL (25 units)
10 mg	2 mL	5 mg/mL (5000 mcg/mL)	0.05 mL (5 units)
10 mg	4 mL	2.5 mg/mL (2500 mcg/mL)	0.10 mL (10 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Upright. Protected from light. Use within 4–6 weeks.

- *For oral use: dissolve in plain sterile water (not BAC water). Drink within 30 minutes. Do not store oral solution.*
- *BPC-157 is one of the most stable peptides post-reconstitution. 4–6 weeks is conservative; many researchers report 8 weeks with proper technique.*
- *Oral and subcutaneous protocols are distinct — oral requires no needles and uses sterile water.*

TB-500 — Thymosin Beta-4 Fragment

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm to room temperature (15 min). Do not open while cold.
2	Alcohol-wipe both stoppers. Air-dry 30 seconds.
3	Draw 1–2 mL BAC water into a 3 mL syringe (see concentration table).
4	Inject slowly down the inside glass wall — never spray directly on powder.
5	Roll gently between palms 60–90 seconds. TB-500 dissolves readily.
6	Solution should be clear and colorless. Label and refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 2.5 mg dose
5 mg	1 mL	5 mg/mL	0.50 mL (50 units)
5 mg	2 mL	2.5 mg/mL	1.00 mL (100 units)
2 mg	1 mL	2 mg/mL	1.25 mL per 2.5 mg
2 mg	2 mL	1 mg/mL	2.50 mL per 2.5 mg

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *TB-500 dissolves easily and is very forgiving. Standard BAC water protocol works every time.*
- *Often stacked with BPC-157 — can reconstitute both in the same timeframe for a healing protocol.*

KPV — Tripeptide (Lys-Pro-Val)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg, 50 mg	**BAC Water (SC) or Sterile Water (oral)**	**4 weeks SC / Immediate oral**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Determine route: SC injection or oral (gut protocol). Solvent differs.
2	For SC: allow vial to warm 15 min, alcohol-wipe stopper, inject 1–2 mL BAC water slowly down glass wall.
3	For oral: use plain sterile water (NOT BAC water — benzyl alcohol is not intended for consumption). Dissolve 50 mg in 5–10 mL sterile water.
4	Roll gently 60 seconds. KPV dissolves readily in both solvents.

5 SC: label, refrigerate, use within 4 weeks. Oral: drink entire solution immediately. Discard any remainder.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Notes
5 mg	1 mL	5 mg/mL	SC protocol — 200–600 mcg doses (0.04–0.12 mL)
5 mg	2 mL	2.5 mg/mL	SC protocol — 200–600 mcg doses (0.08–0.24 mL)
50 mg	5 mL sterile water	10 mg/mL	Oral protocol — drink entire dose immediately
50 mg	10 mL sterile water	5 mg/mL	Oral protocol — more dilute, easier to swallow

STORAGE AFTER RECONSTITUTION

SC vials: refrigerate, use within 4 weeks. Oral solution: use immediately, no storage.

■ **Do not use BAC water for oral administration — benzyl alcohol is not for ingestion.**

- *KPV binds the MC1R receptor on intestinal epithelial and immune cells. Oral and SC routes have different absorption profiles.*
- *Gut protocol research uses high oral doses (50–200 mg). SC research uses lower doses (200–600 mcg).*

GROWTH HORMONE SECRETAGOGUES

CJC-1295 DAC — Growth Hormone Releasing Hormone Analogue

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm to room temperature 15–20 min.
2	Alcohol-wipe both stoppers. Air-dry.
3	Draw 1–2 mL BAC water into syringe.
4	Inject slowly down the glass wall. CJC-1295 DAC dissolves readily.
5	Roll gently 60 seconds. Clear, colorless solution expected.
6	Label with name, concentration, and date. Refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 1 mg dose (weekly)
2 mg	1 mL	2 mg/mL	0.50 mL (50 units)
2 mg	2 mL	1 mg/mL	1.00 mL (100 units)
5 mg	2 mL	2.5 mg/mL	0.40 mL (40 units)
5 mg	5 mL	1 mg/mL	1.00 mL (100 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *DAC (Drug Affinity Complex) gives CJC-1295 a 7–8 day half-life, so weekly dosing is standard.*
- *One vial reconstituted covers 2–5 weeks depending on dose. Stable enough to make weekly dosing practical.*

CJC-1295 No DAC (Mod GRF 1-29)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.

2	Draw 1–2 mL BAC water. Inject slowly down the glass wall.
3	Roll gently 60 seconds. Dissolves readily.
4	Label and refrigerate. Use within 4–6 weeks.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 100 mcg dose
2 mg	2 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)
2 mg	1 mL	2 mg/mL (2000 mcg/mL)	0.05 mL (5 units)
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *No DAC = short half-life (~30 min). Dosed 2–3x/day or pre-workout. Often pre-mixed with Ipamorelin.*
- *When stacking with Ipamorelin: can reconstitute both in the same vial if you know your dose ratio.*

Ipamorelin — Selective Growth Hormone Secretagogue

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 1–2 mL BAC water. Inject slowly down the glass wall.
3	Roll gently 60 seconds. Dissolves immediately and cleanly.
4	Label and refrigerate. Use within 4–6 weeks.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 200 mcg dose
2 mg	1 mL	2 mg/mL (2000 mcg/mL)	0.10 mL (10 units)
2 mg	2 mL	1 mg/mL (1000 mcg/mL)	0.20 mL (20 units)
5 mg	2 mL	2.5 mg/mL (2500 mcg/mL)	0.08 mL (8 units)
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.20 mL (20 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *Ipamorelin is the cleanest GHRP — no cortisol, no prolactin, no appetite surge. Ideal for GH pulse stacking.*
- *Pre-sleep dosing is the most common protocol: matches the natural nocturnal GH pulse.*

Sermorelin — GHRH Analogue (First 29 AA)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
3 mg, 6 mg, 9 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15–20 min. Alcohol-wipe stoppers.
2. Draw 1–2 mL BAC water (see table). Inject slowly down glass wall.
3. Roll gently 60–90 seconds. Clear solution expected.
4. Label with date and concentration. Refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 300 mcg dose
3 mg	1 mL	3 mg/mL (3000 mcg/mL)	0.10 mL (10 units)
3 mg	3 mL	1 mg/mL (1000 mcg/mL)	0.30 mL (30 units)
6 mg	2 mL	3 mg/mL (3000 mcg/mL)	0.10 mL (10 units)
9 mg	3 mL	3 mg/mL (3000 mcg/mL)	0.10 mL (10 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *Sermorelin has a short half-life (~10–12 min). Pre-sleep SC dosing is standard.*
- *Often stacked with Ipamorelin for synergistic GH pulse. Safer hormonal profile than exogenous HGH.*

GHRP-2 — Growth Hormone Releasing Peptide 2

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg	**BAC Water (preferred) or 0.6% Acetic Acid if needed**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.
2. Attempt BAC water first: draw 1–2 mL, inject slowly down glass wall, roll gently 2–5 min.
3. If cloudiness persists after 5 min of rolling: discard BAC water attempt. Use 0.6% acetic acid instead.
4. Acetic acid protocol: draw 0.5 mL 0.6% acetic acid, inject slowly, roll 5–10 min until clear.
5. Once dissolved in acetic acid, add remaining BAC water to reach target volume.

6	Label and refrigerate immediately. Solution should be clear.

CONCENTRATION REFERENCE

Vial Size	Total Volume	Concentration	Volume per 100 mcg dose
5 mg	2 mL	2.5 mg/mL	0.04 mL (4 units)
5 mg	5 mL	1 mg/mL	0.10 mL (10 units)
10 mg	2 mL	5 mg/mL	0.02 mL (2 units)
10 mg	5 mL	2 mg/mL	0.05 mL (5 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 3–4 weeks.

■ **GHRP-2 causes significant cortisol and prolactin spike vs Ipamorelin. Note if sensitive.**

• *GHRP-2 causes strong hunger signal. Dose 30–60 min before meals or fasted for GH pulse.*

• *Stronger GH release than Ipamorelin but less selective. Good for bulking protocols.*

GHRP-6 — Growth Hormone Releasing Peptide 6

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg	**BAC Water (preferred) or 0.6% Acetic Acid if needed**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Attempt BAC water first: 1–2 mL, inject slowly down glass wall, roll 2–5 min.
3	If cloudy after 5 min, switch to 0.6% acetic acid: 0.5 mL first, roll until clear, then add BAC water.
4	Label and refrigerate. Clear, colorless solution expected.

CONCENTRATION REFERENCE

Vial Size	Total Volume	Concentration	Volume per 100 mcg dose
5 mg	2 mL	2.5 mg/mL (2500 mcg/mL)	0.04 mL (4 units)
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)
10 mg	5 mL	2 mg/mL (2000 mcg/mL)	0.05 mL (5 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 3–4 weeks.

■ **GHRP-6 causes the strongest hunger signal of all GHRPs — hunger spikes 20–40 min post-injection.**

• *GHRP-6 is the original GHRP. Powerful GH release but the hunger effect is pronounced.*

• *Often used in bulking/mass-building protocols where the hunger effect is a feature, not a bug.*

Hexarelin — GHRP Analogue (Most Potent GHRP)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg, 5 mg	**0.6% Acetic Acid FIRST, then dilute with BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 0.5 mL of 0.6% acetic acid into a syringe.
3	Inject acetic acid SLOWLY down the glass wall. Do not spray on powder.
4	Roll vial gently for 5–10 minutes. Hexarelin may be slow to dissolve — patience required.
5	Once dissolved (clear solution), draw up the desired additional BAC water to reach target volume.
6	Inject BAC water slowly into the same vial. Roll gently again 60 seconds.
7	Final solution should be clear. Label and refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	0.6% AA	BAC Water Added	Concentration	Vol per 200 mcg dose
2 mg	0.5 mL	0.5–1 mL	1–2 mg/mL	0.10–0.20 mL
5 mg	0.5 mL	1.5–2 mL	2–2.5 mg/mL	0.08–0.10 mL

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 3–4 weeks.

■ **Do NOT skip the acetic acid step. BAC water alone will cause aggregation and loss of potency.**

■ **Hexarelin causes rapid GH receptor desensitization. Do not use daily for more than 4 weeks without a break.**

• *Most potent GHRP by GH release per mcg. Desensitization is faster than other GHRPs.*

• *Cardioprotective effects in animal research, independent of GH release. Used in heart-focused protocols.*

Tesamorelin — Stabilized GHRH Analogue (FDA-Approved)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
1 mg, 2 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15–20 min. Alcohol-wipe stoppers.
2	Draw 0.5–1 mL BAC water per mg of peptide (see table).
3	Inject slowly down the glass wall. Tesamorelin can sometimes be gel-like — go slowly.
4	If powder is slow to dissolve, warm sealed vial in palm at body temperature for 3–5 min, then roll gently again.
5	Do NOT use acetic acid unless powder completely refuses to dissolve after 15 min of trying.
6	Clear solution expected. Label and refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 2 mg dose
1 mg	1 mL	1 mg/mL	2.00 mL (200 units) — use two draws
2 mg	1 mL	2 mg/mL	1.00 mL (100 units)
2 mg	2 mL	1 mg/mL	2.00 mL (200 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

■ **Some batches of Tesamorelin can form a gel. Do not panic — body temperature warming resolves it.**

• *FDA-approved for HIV-associated lipodystrophy (Egrifta). Research-grade versions follow the same reconstitution protocol as the pharmaceutical.*

• *Daily SC injection in abdomen. Tesamorelin preferentially reduces visceral adipose tissue (belly fat).*

IGF & GH PEPTIDES

IGF-1 LR3 — Insulin-Like Growth Factor 1 Long R3

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
1 mg, 2 mg	**0.6% Acetic Acid FIRST — mandatory**	**2–3 weeks refrigerated / 3 months frozen**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15–20 min. Alcohol-wipe stoppers.
2	Draw 0.3–0.5 mL of 0.6% acetic acid into a syringe.
3	Inject acetic acid SLOWLY down the inside glass wall of the IGF-1 vial.
4	Do NOT inject directly onto the powder — inject so the liquid runs down the wall and contacts the powder gently.
5	Roll vial slowly between palms for 10–15 minutes. IGF-1 LR3 can be slow to dissolve. Do not rush.
6	If still not fully dissolved after 15 min: warm sealed vial at 37°C (body temperature or warm water bath) for 5 min, then roll again.
7	Once fully dissolved, add BAC water to reach target final volume (see table). Inject BAC water slowly.
8	Roll gently 60 seconds to mix. Solution should be clear and colorless.
9	Aliquot into single-use amounts immediately after reconstitution if possible, and freeze aliquots.
10	Label all vials with compound name, date, and concentration.

CONCENTRATION REFERENCE

Vial Size	0.6% AA First	BAC Water to Add	Final Concentration	Vol per 100 mcg dose
1 mg	0.3 mL	0.7 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)
1 mg	0.5 mL	0.5 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)
2 mg	0.5 mL	1.5 mL	1 mg/mL (1000 mcg/mL)	0.10 mL (10 units)
2 mg	0.5 mL	0.5 mL	2 mg/mL (2000 mcg/mL)	0.05 mL (5 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 2–3 weeks. Or aliquot and freeze (-20°C) for up to 3 months.

■ **NEVER use BAC water alone for IGF-1. It will aggregate immediately and potency will be destroyed.**

■ **0.6% acetic acid is not optional. It is required to dissolve IGF-1 correctly.**

■ IGF-1 has profound anabolic and hypoglycemic potential. Research use only — handle with care.

• *LR3 variant has a 20–30 hour half-life vs 15–20 min for native IGF-1. Once-daily dosing is typical in research.*

• *Hypoglycemia risk: many researchers inject immediately post-workout with food nearby.*

• *Aliquoting before storage is strongly recommended due to the short refrigerated shelf life.*

MGF — Mechano Growth Factor (IGF-1 Splice Variant)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
2 mg	0.6% Acetic Acid FIRST — mandatory	2 weeks refrigerated / 3 months frozen	Clear, colorless solution

RECONSTITUTION STEPS

1. Allow vial to warm 15–20 min. Alcohol-wipe stoppers.
2. Draw 0.3–0.5 mL of 0.6% acetic acid.
3. Inject SLOWLY down inside glass wall. Roll gently for 10–15 min.
4. If needed, warm sealed vial in palm for 5 min and continue rolling.
5. Once dissolved, add BAC water to reach target volume. Roll 60 seconds.
6. Aliquot into single-use amounts and freeze. MGF has a short refrigerated shelf life.
7. Label all vials with name, date, concentration.

CONCENTRATION REFERENCE

Vial Size	0.6% AA First	BAC Water to Add	Final Concentration	Vol per 200 mcg dose
2 mg	0.3 mL	0.7 mL	2 mg/mL (2000 mcg/mL)	0.10 mL (10 units)
2 mg	0.5 mL	1.5 mL	1 mg/mL (1000 mcg/mL)	0.20 mL (20 units)

STORAGE AFTER RECONSTITUTION

Refrigerate: use within 2 weeks. Freeze aliquots at -20°C for up to 3 months.

■ Same critical acetic acid requirement as IGF-1. BAC water alone = aggregation = wasted vial.

■ MGF has a very short half-life (~ minutes). Timing to post-workout is critical in research protocols.

• *MGF (Mechano Growth Factor) is a splice variant of IGF-1 produced in muscle in response to mechanical damage.*

• *PEG-MGF (pegylated) has a much longer half-life and does NOT require acetic acid — different compound entirely.*

AOD-9604 — HGH Fragment for Fat Metabolism

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE

2 mg, 5 mg	**0.6% Acetic Acid (1:1) + BAC Water — prevents gelling**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 0.5–1 mL of 0.6% acetic acid (1:1 ratio with BAC water you plan to add).
3	Inject acetic acid SLOWLY down the inside glass wall. Do not spray onto the powder.
4	Roll gently 2–3 minutes. AOD-9604 gels in BAC water alone — acetic acid prevents this.
5	Once dissolved, add BAC water slowly down the glass wall to reach your target final volume.
6	Roll gently 60 seconds to mix. Solution should be clear and colorless.
7	If gelling still occurs after adding BAC water: draw additional BAC water into the vial and continue rolling. Do not discard.
8	Label vial with name, date, and concentration. Refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	0.6% AA First	BAC Water	Final Vol	Concentration	Vol per 500 mcg dose
2 mg	0.5 mL	0.5 mL	1 mL	2 mg/mL (2000 mcg/mL)	0.25 mL (25 units)
2 mg	0.5 mL	1.5 mL	2 mL	1 mg/mL (1000 mcg/mL)	0.50 mL (50 units)
5 mg	0.5 mL	1.5 mL	2 mL	2.5 mg/mL (2500 mcg/mL)	0.20 mL (20 units)
5 mg	1 mL	4 mL	5 mL	1 mg/mL (1000 mcg/mL)	0.50 mL (50 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

■ AOD-9604 WILL gel if reconstituted in BAC water alone. Always use acetic acid first (1:1 ratio minimum).

■ If gelling still occurs after adding BAC water: draw more BAC water into the vial and roll until the gel dissolves. Do not discard the vial.

• *AOD-9604 is the C-terminal fragment (176–191) of HGH. Stimulates fat breakdown without IGF-1 axis activation.*

• *No significant effect on blood sugar. Safer hormonal profile than full HGH for fat loss research.*

GLP-1 / GIP METABOLIC PEPTIDES

Semaglutide — GLP-1 Receptor Agonist

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
3 mg, 5 mg, 10 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15–20 min. Alcohol-wipe stoppers.
2	Determine your target concentration based on titration schedule (see table).
3	Draw the appropriate volume of BAC water. Inject slowly down glass wall.
4	Roll very gently 60–90 seconds. Semaglutide dissolves readily.
5	Solution should be clear and colorless. Label with name, date, concentration, and weekly dose.
6	Refrigerate immediately. One vial typically covers multiple weeks at low doses.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Draw per 0.25 mg dose	Draw per 0.5 mg dose	Draw per 1 mg dose
3 mg	1.5 mL	2 mg/mL	0.125 mL (12.5 units)	0.25 mL (25 units)	0.50 mL (50 units)
5 mg	2.5 mL	2 mg/mL	0.125 mL (12.5 units)	0.25 mL (25 units)	0.50 mL (50 units)
5 mg	5 mL	1 mg/mL	0.25 mL (25 units)	0.50 mL (50 units)	1.00 mL (100 units)
10 mg	5 mL	2 mg/mL	0.125 mL	0.25 mL	0.50 mL

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks. Do NOT freeze reconstituted semaglutide.

■ **Do NOT freeze reconstituted semaglutide — it can cause aggregation and loss of potency.**

• *Half-life ~7 days. Weekly SC injection is the standard protocol.*

• *Standard titration: 0.25 mg/week x 4 weeks → 0.5 mg/week x 4 weeks → 1 mg/week. Slower titration reduces GI side effects.*

• *Abdominal SC injection (rotate sites). Thigh and upper arm are alternatives.*

Tirzepatide — GLP-1/GIP Dual Agonist

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE

5 mg, 10 mg, 15 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1 Allow vial to warm 15–20 min. Alcohol-wipe stoppers.

2 Determine target concentration based on titration stage.

3 Draw BAC water per table. Inject slowly down glass wall.

4 Roll gently 60–90 seconds. Dissolves readily.

5 Label: 'Tirzepatide — [date] — [mg/mL]'. Refrigerate immediately.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Draw per 2.5 mg dose	Draw per 5 mg dose
5 mg	2.5 mL	2 mg/mL	1.25 mL	2.50 mL
5 mg	1 mL	5 mg/mL	0.50 mL (50 units)	1.00 mL (100 units)
10 mg	2 mL	5 mg/mL	0.50 mL	1.00 mL
15 mg	3 mL	5 mg/mL	0.50 mL	1.00 mL

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

- *Standard titration: 2.5 mg/week x 4 weeks → 5 mg/week x 4 weeks → 7.5 mg → 10 mg → 12.5 mg → 15 mg. Slow titration is key to GI tolerance.*
- *GIP component adds superior weight loss vs GLP-1 alone. Clinical trials show 20–22% body weight reduction.*
- *Weekly SC injection. Same injection sites as semaglutide.*

Retatrutide — GLP-1/GIP/Glucagon Triple Agonist

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1 Allow vial to warm 15–20 min. Alcohol-wipe stoppers.

2 Draw BAC water per table. Inject slowly down glass wall.

3 Roll gently 60 seconds. Dissolves readily.

4 Label and refrigerate. Clear, colorless solution.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Draw per 2 mg dose	Draw per 4 mg dose

5 mg	2.5 mL	2 mg/mL	1.00 mL (100 units)	2.00 mL
5 mg	1 mL	5 mg/mL	0.40 mL (40 units)	0.80 mL
10 mg	2 mL	5 mg/mL	0.40 mL	0.80 mL

STORAGE AFTER RECONSTITUTION
Refrigerate at 2–8°C. Use within 4–6 weeks.

• *Triple agonist. Glucagon component adds thermogenic effect and enhanced fat oxidation on top of GLP-1/GIP.*

• *Phase 3 trials show up to 24% body weight loss. Not yet FDA-approved. Research use only.*

• *Titrate slowly. Start 2 mg/week. GI adaptation is critical.*

SKIN, TANNING & LIBIDO PEPTIDES

PT-141 (Bremelanotide) — Melanocortin Sexual Peptide

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
10 mg	**BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 1–2 mL BAC water. Inject slowly down glass wall.
3	Roll gently 60 seconds. Dissolves readily.
4	Label and refrigerate. Protect from light — use amber vial or wrap in foil.
5	For nasal spray: dissolve 10 mg in 10 mL BAC water = 1 mg/mL solution. Transfer to nasal atomizer.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 1 mg dose	Volume per 1.75 mg dose
10 mg	2 mL	5 mg/mL (5000 mcg/mL)	0.20 mL (20 units)	0.35 mL (35 units)
10 mg	5 mL	2 mg/mL (2000 mcg/mL)	0.50 mL (50 units)	0.875 mL
10 mg	10 mL	1 mg/mL (nasal spray)	1.00 mL (nasal spray)	—

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Protect from light. Use within 3–4 weeks.

- *Onset 45–60 min post-SC injection. Duration 6–12 hours. Plan accordingly.*
- *FDA-approved version (Vyleesi) is 1.75 mg SC. Research dosing commonly starts at 0.5–1 mg.*
- *Nasal spray route has faster onset but requires precise atomizer technique.*
- *Transient nausea and flushing are common side effects, especially at higher doses.*

Melanotan II (MT-II) — Tanning & Melanocortin Peptide

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
10 mg	**BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.
2. Draw 1–2 mL BAC water. Inject slowly down glass wall.
3. Roll gently 60 seconds. Dissolves readily.
4. Protect from light immediately — transfer to amber vial or wrap in aluminum foil.
5. Label and refrigerate. Use within 3–4 weeks.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 0.5 mg dose	Volume per 1 mg dose
10 mg	2 mL	5 mg/mL (5000 mcg/mL)	0.10 mL (10 units)	0.20 mL (20 units)
10 mg	5 mL	2 mg/mL (2000 mcg/mL)	0.25 mL (25 units)	0.50 mL (50 units)
10 mg	10 mL	1 mg/mL (1000 mcg/mL)	0.50 mL (50 units)	1.00 mL (100 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Protect from light in amber vial. Use within 3–4 weeks.

■ **MT-II is extremely light-sensitive. Store in amber vial wrapped in foil. Never expose to sunlight.**

■ **MT-II activates melanocytes. Watch for changes in moles/nevi — a known research concern.**

• *Nausea is very common on first injection, especially at doses above 0.25 mg. Start low.*

• *Tanning loading phase: 0.25–0.5 mg/day for 2–3 weeks, with UV exposure. Maintenance: 0.5–1 mg every few days.*

• *Also causes libido effects (same MC receptor pathway as PT-141).*

GHK-Cu — Copper Peptide (Glycine-Histidine-Lysine-Copper)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
50 mg, 100 mg, 200 mg	**BAC Water (SC) or Sterile Water / PG blend (topical)**	**2–3 weeks SC (refrigerated)**	**BLUE solution (copper complex) — NORMAL. Not purple. Not colorless. Blue.**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.
2. For SC: draw at least 3 mL BAC water — extra dilution significantly reduces injection site stinging.
3. Inject BAC water SLOWLY down the inside glass wall. Do not spray onto the powder.
4. Roll gently 60–90 seconds. The solution will turn BLUE — this is NORMAL and correct.
5. If injection site stinging still occurs: next time draw additional BAC water into syringe before injecting to dilute further.

6 For topical: dissolve in sterile water or propylene glycol (PG) 20% / sterile water 80% blend.

7 Label and refrigerate. The blue color is the copper complex and does not indicate degradation.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Vol per 1 mg dose (SC)	Stinging Risk
50 mg	50 mL	1 mg/mL	1.00 mL (100 units)	Low — recommended
50 mg	25 mL	2 mg/mL	0.50 mL (50 units)	Moderate
100 mg	100 mL	1 mg/mL	1.00 mL (100 units)	Low — recommended
Topical	PG 20% / water 80%	Variable	Apply to skin	N/A

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Protect from light. SC: 2–3 weeks. Topical: 4–6 weeks.

■ GHK-Cu solution is BLUE — not purple, not colorless. Blue = copper complex formed correctly.

■ A colorless GHK-Cu solution may mean the copper was lost. Blue = correct.

■ Always use at least 3 mL BAC water to reduce stinging. If stinging still occurs, draw more BAC water into syringe before injecting.

• GHK-Cu degrades faster than most peptides due to copper oxidation. Use within 2–3 weeks.

• Topical is the most common route for skin, hair, and wound healing research.

• Lower concentration = less stinging. 3 mL minimum is the most reliable fix for injection site discomfort.

COGNITIVE & NASAL PEPTIDES

Semax — ACTH(4–7) Analogue (Nasal)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
1 mg, 5 mg, 10 mg	**BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.
2. Determine target concentration for nasal spray (1 mg/mL is standard for most protocols).
3. Draw BAC water per table. Inject slowly down glass wall.
4. Roll gently 60 seconds. Dissolves readily.
5. Transfer to nasal spray atomizer. Label and refrigerate.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Drops per 100 mcg dose (Nasal)
1 mg	1 mL	1 mg/mL (1000 mcg/mL)	0.10 mL ~ 2 drops per nostril
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.10 mL ~ 2 drops per nostril
5 mg	2.5 mL	2 mg/mL (2000 mcg/mL)	0.05 mL ~ 1 drop per nostril

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C in atomizer or original vial. Use within 3–4 weeks.

- *Nasal route is standard. Each nostril receives 1–2 drops (50–100 mcg). Sniff gently after application.*
- *Na-Semax (N-acetylated) has the same reconstitution protocol but is considered more potent per mcg.*
- *BDNF-boosting effects are the primary research focus. Cognitive enhancement, neuroprotection, anxiety reduction.*

Selank — Anxiolytic Peptide (Nasal)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg	**BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.

2	Draw BAC water per table. Standard is 1 mg/mL for nasal use.
3	Inject slowly down glass wall. Roll gently 60 seconds.
4	Transfer to nasal spray atomizer. Label and refrigerate.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Drops per 250 mcg dose (Nasal)
5 mg	5 mL	1 mg/mL (1000 mcg/mL)	0.25 mL ~ 5 drops
5 mg	2.5 mL	2 mg/mL (2000 mcg/mL)	0.125 mL ~ 2–3 drops
10 mg	10 mL	1 mg/mL (1000 mcg/mL)	0.25 mL ~ 5 drops

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C in atomizer. Use within 3–4 weeks.

- *Selank modulates the GABAergic system. Research focus: anxiety, stress, cognitive function.*
- *Na-Selank Amidate is the longer-acting variant. Same reconstitution protocol.*
- *No sedation reported in research at typical doses. Anxiolytic without drowsiness is the key profile.*

LONGEVITY & ANTI-AGING PEPTIDES

Epitalon (Epithalon) — Telomere Peptide

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
10 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 1 mL BAC water (for 10 mg/mL) or 2 mL (for 5 mg/mL).
3	Inject slowly down glass wall. Roll gently 60 seconds.
4	Epitalon dissolves readily. Clear, colorless solution.
5	Label and refrigerate. Typical Khavinson protocol: 10-day course twice per year.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 10 mg dose	Volume per 5 mg dose
10 mg	1 mL	10 mg/mL	1.00 mL (100 units)	0.50 mL (50 units)
10 mg	2 mL	5 mg/mL	2.00 mL	1.00 mL (100 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

• *Original Khavinson protocol: 10 mg/day SC for 10 consecutive days. Repeat 2x/year.*

• *Epitalon stimulates telomerase and pineal gland function in research models.*

• *Na-Epitalon (N-acetylated) has the same reconstitution protocol.*

MOTS-C — Mitochondrial Peptide (Metabolic & Longevity)

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
5 mg, 10 mg	**BAC Water**	**4–6 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw 1–2 mL BAC water. Inject slowly down glass wall.
3	Roll gently 60 seconds. Dissolves readily.

4	Label and refrigerate. Clear, colorless solution.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 5 mg dose	Volume per 10 mg dose
5 mg	1 mL	5 mg/mL	1.00 mL (100 units)	—
5 mg	2.5 mL	2 mg/mL	2.50 mL	—
10 mg	2 mL	5 mg/mL	1.00 mL (100 units)	2.00 mL

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 4–6 weeks.

• *MOTS-C is a mitochondrial-derived peptide. Exercise timing (pre-workout) enhances metabolic effects in research.*

• *Emerging longevity research target. AMPK activation is the primary mechanism.*

SS-31 (Elamipretide) — Mitochondrial Membrane Protector

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
10 mg, 20 mg	**BAC Water**	**2–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1	Allow vial to warm 15 min. Alcohol-wipe stoppers.
2	Draw BAC water per table. Work quickly to minimize air/light exposure.
3	Inject slowly down glass wall. Roll gently 60 seconds.
4	Transfer to amber vial or wrap in foil immediately. SS-31 is oxidation-sensitive.
5	Label and refrigerate. Solution should be clear and colorless.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 10 mg dose	Volume per 20 mg dose
10 mg	1 mL	10 mg/mL	1.00 mL (100 units)	—
10 mg	2 mL	5 mg/mL	2.00 mL	—
20 mg	2 mL	10 mg/mL	1.00 mL (100 units)	2.00 mL

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Protect from light in amber vial. Use within 2–4 weeks.

■ **Yellow color after reconstitution = oxidation = discard. Clear and colorless only.**

■ **Protect from light. Oxidation can occur within hours if exposed to bright light.**

• *SS-31 is in Phase 2 clinical trials for heart failure (EMPA-HEART) and Barth syndrome.*

• *Targets cardiolipin on the inner mitochondrial membrane. Used in cardiac and anti-aging research.*

Thymosin Alpha-1 (Ta1) — Immune Modulator

COMMON VIAL SIZES	SOLVENT	SHELF LIFE (RECON)	APPEARANCE
1.6 mg	**BAC Water**	**3–4 weeks (refrigerated)**	**Clear, colorless solution**

RECONSTITUTION STEPS

1. Allow vial to warm 15 min. Alcohol-wipe stoppers.
2. Draw exactly 1 mL BAC water for the standard 1.6 mg vial.
3. Inject slowly down glass wall. Roll gently 60 seconds.
4. Dissolves readily. Label and refrigerate.

CONCENTRATION REFERENCE

Vial Size	BAC Water Added	Concentration	Volume per 1.6 mg dose
1.6 mg	1 mL	1.6 mg/mL	1.00 mL (100 units) — entire vial
1.6 mg	0.8 mL	2 mg/mL	0.80 mL (80 units)

STORAGE AFTER RECONSTITUTION

Refrigerate at 2–8°C. Use within 3–4 weeks.

• *Sold commercially as Zadaxin (FDA-approved in 35+ countries). Research vials are typically 1.6 mg to match Zadaxin dosing.*

• *Typical protocol: 1.6 mg SC twice weekly. Used in immune modulation, chronic infection, and cancer support research.*

CONCENTRATION MATH REFERENCE

How to Calculate Your Draw Volume

The formula for calculating how many mL to draw for a given dose is:

Draw Volume (mL) = Desired Dose (mg) ÷ Concentration (mg/mL)

Unit conversion for insulin syringes (U-100):

U-100 insulin syringes are calibrated for 100 units per mL. 1 mL = 100 units. So 0.10 mL = 10 units. 0.25 mL = 25 units.

Example calculations:

BPC-157 — 250 mcg dose, vial reconstituted at 2.5 mg/mL

→ 250 mcg = 0.25 mg. Draw = 0.25 ÷ 2.5 = 0.10 mL = 10 units on U-100 syringe.

Semaglutide — 0.5 mg dose, reconstituted at 2 mg/mL

→ Draw = 0.5 ÷ 2 = 0.25 mL = 25 units on U-100 syringe.

Ipamorelin — 200 mcg dose, reconstituted at 1 mg/mL

→ 200 mcg = 0.2 mg. Draw = 0.2 ÷ 1 = 0.20 mL = 20 units on U-100 syringe.

IGF-1 LR3 — 100 mcg dose, reconstituted at 1 mg/mL

→ 100 mcg = 0.1 mg. Draw = 0.1 ÷ 1 = 0.10 mL = 10 units on U-100 syringe.

Tirzepatide — 5 mg dose, reconstituted at 5 mg/mL

→ Draw = 5 ÷ 5 = 1.00 mL = 100 units on U-100 syringe.